Small cameras, big costs

Camera-to-inference latency on NVIDIA Jetson AGX Xavier with a USB 2.0 UVC camera: zero-copy against a copy path, measured stage by stage

axon SDK (github.com/ohora23/Axon_SDK_for_PhysicalAI) · measured 2026-11-11 on hermes-head · code `examples/jetson_e2e`, runner `instrumentation/jetson/e2e/run_e2e.sh`, raw data `instrumentation/jetson/e2e/results/` (PR #60) · paper PR #61

Abstract. Robot hands and grippers often carry small UVC cameras, such as USB 2.0, 640×480, 30 fps devices. We measured where the time goes between such a camera and a neural network on an embedded GPU, frame by frame, for two pipelines that differ only in how the frame reaches the GPU: *zero-copy*, where the driver writes the frame into a buffer the GPU can read directly (axon `NvBufSurface` pool), and *copy*, the usual path (driver buffer → `memcpy` into shared memory → host-to-device copy). On a Jetson AGX Xavier with a ResNet-18 TensorRT engine, zero-copy saves the host-to-device copy, and nothing else: 0.43 ms per frame at the default power settings and 0.20 ms with locked clocks (4 % and 10 % of receive-to-inference). Two other costs are larger: the first GPU operation after the idle gap between 30 fps frames (0.9–1.1 ms at default clocks; cause not isolated) and the interval from the frame's V4L2 timestamp to its availability, which covers the USB 2.0 transfer (about 32 ms). Our first reading of the data attributed the wake-up cost to slow GPU reads of the camera buffer; a control experiment refuted it, and we report both.

Key findings

- **Zero-copy removes the H2D copy and nothing else**: 423 → 0 µs (default clocks), 184 → 0 µs (`jetson_clocks`) per 614 KB frame. The pre-processing kernel runs equally fast on the camera buffer and on `cudaMalloc` memory (234 vs 237 µs; 60 vs 62 µs).
- **The first GPU operation after idle costs more than the copy.** At 30 fps the GPU idles about 30 ms per frame; the first GPU operation afterwards pays 0.9–1.1 ms (default clocks) or 0.3–0.36 ms (locked clocks). Clock ramp, power gating and the driver submit path were not separated.
- **The camera dominates end to end.** 32–36 ms pass between the frame's V4L2 timestamp (first USB packet) and the frame being available, for both pipelines (USB transfer plus driver dequeue).
- **The clock policy sets inference time**: 10.8 ms at the default 30 W mode against 1.7 ms with `jetson_clocks` (GPU and memory clocks locked), for the same engine.
- For a hand with several cameras the copy cost scales with the camera count (0.2–0.4 ms each, per frame), while the camera and wake-up costs do not shrink with zero-copy.

1. Why small cameras

Manipulation increasingly relies on cameras mounted on the hand: wrist and palm cameras give close-range views of contact that a head camera cannot. These cameras are small and cheap; many are UVC devices on USB 2.0, delivering YUYV or MJPEG at VGA resolution and 30 fps. The compute that consumes them is often an embedded GPU on the robot (Jetson class), shared with control and other perception. Two questions follow for a system designer: how

much of the camera-to-decision latency can the software path still remove, and where does the rest go?

axon is an inter-process transport that hands frames between processes without copying the payload: producers write into a shared buffer pool and publish only a small descriptor. On Jetson it can pool `NvBufSurface` buffers, which a UVC driver can fill directly and which CUDA can map, so a consumer process gets the frame as a GPU pointer with no copy in user space (§6 of `docs/hardware-verification.md`). This paper measures what that buys in a complete camera → pre-processing → inference loop.

2. Setup

Item	Configuration
Board	NVIDIA Jetson AGX Xavier, L4T R35.6 (JetPack 5.1), CUDA 11.4, TensorRT 8.5.2, cuDNN 8.6
Kernel	PREEMPT_RT `5.10.216-rt108-tegra`, CPUs 4–7 isolated (`isolcpus=managed_irq,domain,4-7 irqaffinity=0-3 kthread_cpus=0-3`)
Power	`MODE_30W_ALL` (CPU 1.19 GHz; GPU devfreq 115–905 MHz); a second set with `jetson_clocks` (GPU fixed at 905 MHz, EMC fixed)
Camera	Logitech Webcam Pro 9000 (UVC, USB 2.0 high speed), 640×480 YUYV, 30 fps, 614 KB per frame
Network	ResNet-18 (ONNX model zoo `resnet18-v1-7`), TensorRT FP16 engine, input 1×3×224×224; `trtexec` alone: 1.60 ms median
Placement	consumer (pre-processing + inference) pinned to isolated CPU 4; producer on CPUs 0–3
Sample	per cell 2 runs × 600 timed frames (after 30 warm-up frames) = 1,200 frames; no frame skipped, no seqlock retry

Two pipelines

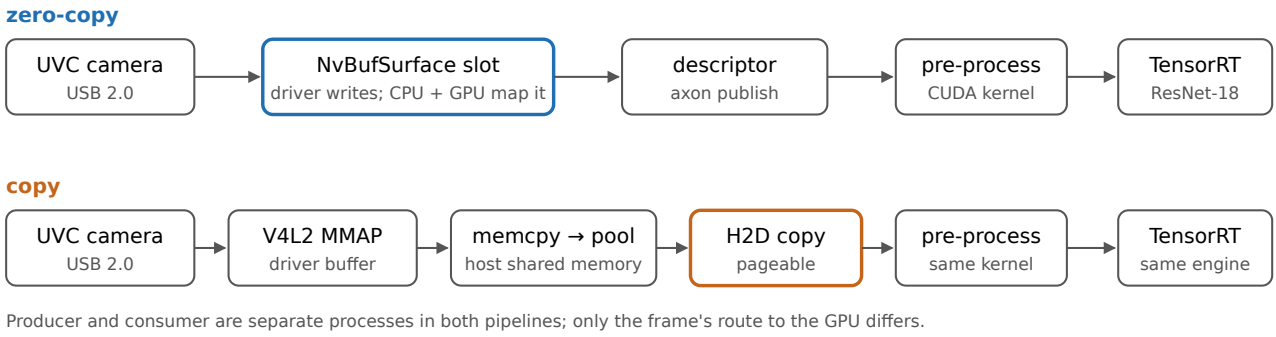


Figure 1. The two pipelines. Both use axon to publish the frame descriptor to the consumer process.

3. Method

Timestamps. Every frame carries six timestamps, all on `CLOCK_MONOTONIC_RAW`: the capture time (the V4L2 buffer timestamp, which uvcvideo takes on `CLOCK_MONOTONIC` when the frame's first USB packet arrives; converted with the offset between the two clocks read at dequeue), publish, received by the consumer, frame available on the GPU, pre-processing done and inference done. Each GPU step is followed by a stream synchronisation, so its time is measured from the host. The capture-time conversion and the derived publish time (`received − staleness`) are accurate to well under 1 µs over these intervals.

Stages. *H2D* is zero for zero-copy (the frame is already GPU-addressable) and a pageable `cudaMemcpyAsync` + sync for copy. The pre-processing kernel converts YUYV to RGB (BT.601),

resizes to 224×224 (nearest) and normalises; inference is `enqueueV3` + sync. *Received* → *inferred* is the consumer-side cost a transport choice can change; *capture* → *inferred* adds the camera and the producer.

GPU warm-up. At 30 fps the GPU is idle for most of each frame period. With `E2E_WARM_GPU=1` the consumer runs a tiny kernel and synchronises after a frame is announced and before it is read, so the first timed GPU step does not also pay for waking the GPU. The warm-up time therefore appears in *publish* → *received*. §5 explains why this matters.

Runs. The two modes alternate within each repetition (zero-copy, copy, zero-copy, copy) so slow drift affects both. Four cells: {default clocks, `jetson_clocks`} × {warm-up, no warm-up}.

4. Results

4.1 With the GPU woken before each frame

stage, µs: p50 (p99)	zero-copy	copy	zero-copy, jetson_clocks	copy, jetson_clocks
capture → publish	32,097 (36,164)	32,204 (36,315)	32,092 (36,155)	32,194 (36,273)
publish → received (incl. warm-up)	1,173 (1,294)	1,181 (1,297)	464 (573)	474 (585)
H2D copy	0 (0)	423 (504)	0 (0)	184 (263)
pre-processing kernel	234 (275)	237 (276)	60 (109)	62 (139)
inference	10,820 (10,892)	10,819 (10,874)	1,684 (1,730)	1,689 (1,752)
received → inferred	**11,056 (11,134)**	**11,486 (11,575)**	**1,746 (1,800)**	**1,947 (2,115)**
capture → inferred	44,361 (48,489)	44,896 (49,048)	34,334 (38,440)	34,646 (38,782)

Table 1. 1,200 frames per cell. Zero-copy saves 430 µs (3.7 %) and 201 µs (10.3 %) of received → inferred; 535 µs and 312 µs of capture → inferred, which also includes the copy producer's `memcpy` (about 0.1 ms, inside capture → publish).

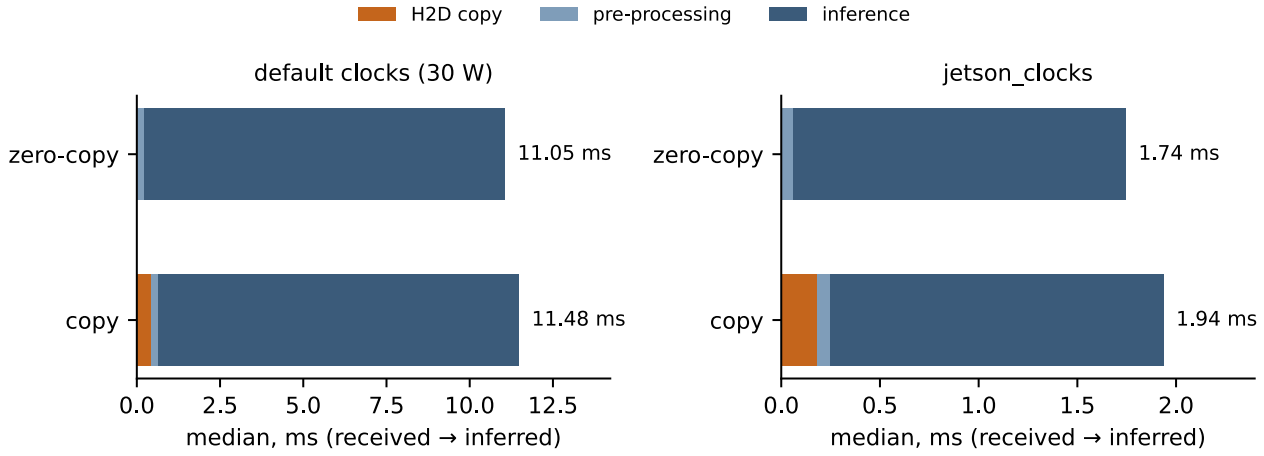


Figure 2. Median received → inferred, split by stage. The only difference between the bars of a panel is the H2D copy.

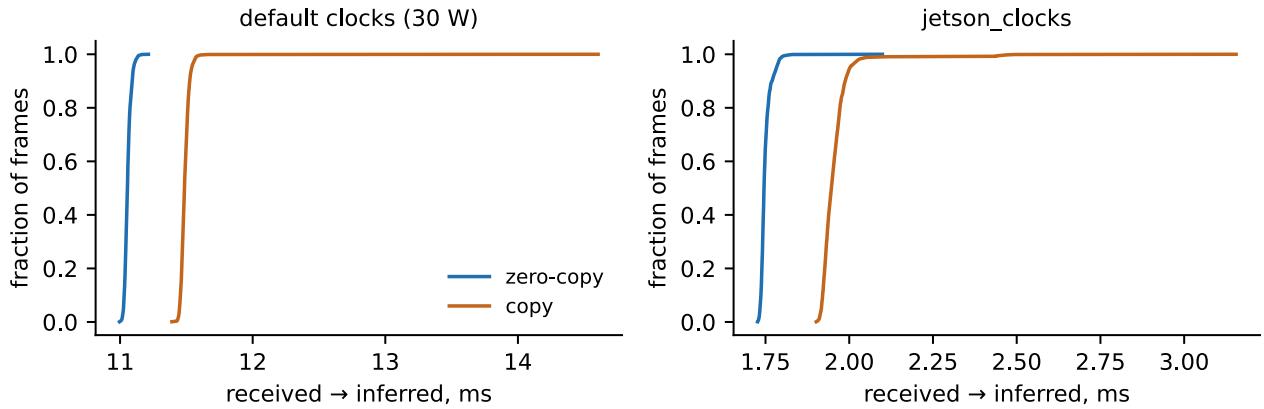


Figure 3. Distribution of received → inferred over 1,200 frames per curve. The curves are narrow (p99 within 0.2 ms of p50); zero-copy is shifted left by the H2D time.

4.2 Without the warm-up

stage, µs: p50	zero-copy	copy	zero-copy, jetson_clocks	copy, jetson_clocks
publish → received	106	109	104	111
H2D copy	0	1,338	0	499
pre-processing kernel	1,164	245	379	65
inference	10,845	10,838	1,693	1,698
received → inferred	**12,010**	**12,427**	**2,077**	**2,271**

Table 2. Same setup, no warm-up (*_nowarm). Whichever GPU step comes first after the idle gap is about 0.9 ms (default) or 0.3 ms (jetson_clocks) slower than in Table 1.

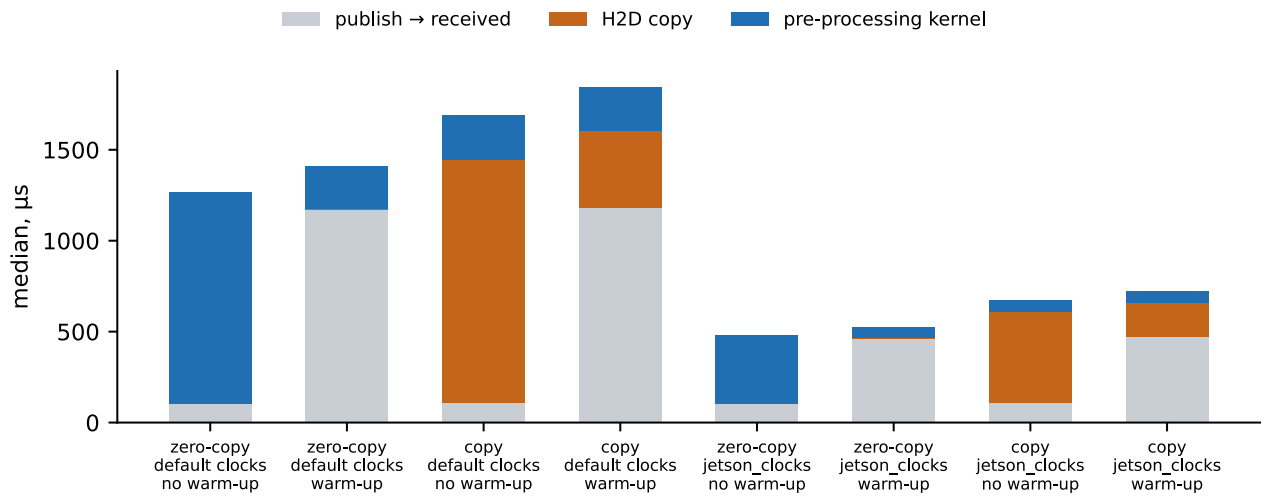


Figure 4. Medians of publish → received, H2D and kernel, with and without the warm-up. The wake-up cost moves: onto the kernel (zero-copy) or the H2D copy (copy) without warm-up, into publish → received with it. Its size is about the same; publish → received also carries the warm-up kernel itself.

5. Analysis

5.1 Zero-copy saves the copy, and nothing else

With the GPU awake, the two pipelines differ in exactly one stage. The H2D copy of a 614 KB frame from pageable memory takes 423 µs at the default clocks (1.45 GB/s) and 184 µs with locked clocks (3.3 GB/s); zero-copy removes it. The pre-processing kernel reads the `NvBufSurface` as fast as a `cudaMalloc` buffer (234 vs 237 µs, 60 vs 62 µs), so no cost appears elsewhere. Inference is identical. The saving is 4 % of received → inferred at the default clocks and 10 % with locked clocks, where inference is shorter and the copy weighs more.

5.2 The first GPU operation after idle costs more than the copy

Without the warm-up (Table 2), the first GPU operation of each frame is slower by about 0.9 ms at the default clocks and 0.32 ms with `jetson_clocks`: the kernel in the zero-copy pipeline, the H2D copy in the copy pipeline. About the same amount reappears in publish → received when the warm-up runs there (1,173 vs 106 µs; 464 vs 104 µs), slightly more because that stage also runs the warm-up kernel. The GPU is idle for about 30 ms of every 33 ms frame period. Locking the clocks shortens but does not remove the effect, so clock ramp alone does not explain it; power gating, the driver submit path and the idle consumer CPU were not separated.

A retracted reading. Our first analysis used only the no-warm-up runs and concluded that the kernel reads the camera buffer 4.7× slower than device memory (1,164 vs 245 µs). A reviewer noticed that in the locked-clock copy runs, the 86 frames whose H2D copy was unusually short

had a kernel time of 376 µs on `cudaMalloc` memory, the same as the zero-copy kernel. That pointed at a first-operation cost, not at the memory. The warm-up experiment confirmed it. For measurements of GPU pipelines at camera rates, the first GPU step after an idle gap should be treated as a separate stage or controlled for.

5.3 The camera dominates

Between the V4L2 timestamp and the frame being ready to publish, 32–36 ms pass in both pipelines (p50 32.1 ms, p99 36.2–36.3 ms). uvcvideo stamps a frame when its first USB packet arrives (`hwtimestamps=0`), so this interval is the USB 2.0 transfer of the frame plus the driver's copy out of the USB buffers and the dequeue; the sensor exposure happened earlier and is not included. The distribution has two modes, near 32 ms and near 36 ms, with a floor at 32.0 ms. The data cannot tell whether the link bandwidth or the camera's own readout pace sets it. Compared with this, received → inferred at locked clocks is 1.75–1.95 ms, or about 2.1–2.3 ms when the first-operation cost is included.

5.4 The clock policy sets inference time

The same engine takes 10.8 ms at the default 30 W mode (a single devfreq read at the start of each run showed the GPU at 115 MHz while idle) and 1.7 ms with `jetson_clocks` (GPU at 905 MHz, memory clock locked too, so the GPU clock alone is not isolated), close to the 1.6 ms that `trtexec` reports when it keeps the GPU busy. For latency-sensitive perception at camera rates, the GPU and memory clock policy matters more than the transport.

5.5 Transport latency

Publish → received without the warm-up, about 106 µs, is mostly the 100 µs polling interval of axon's `wait_for_update` on this build: the JetPack 5 kernel headers lack `futex_waitv`, so the wait falls back to polling. On kernels ≥ 5.16 with matching headers the wait is futex-based.

6. Implications for hand-mounted cameras

Observation	Design implication
Timestamp → available ≈ 32 ms of a ~44 ms (default) or ~34 ms (locked) capture → inferred	Latency targets below ~35 ms likely need a different camera: USB 3 or CSI, a higher frame rate, or, if the link bandwidth is the limit (not established here), lower resolution or MJPEG (decode on Jetson's NVJPG). The transport software cannot recover this part.
Zero-copy saves 0.2–0.4 ms per frame per camera	Worth it per camera, and it adds up: four hand cameras sharing one GPU would save about 0.8–1.7 ms per cycle in H2D time alone (arithmetic, not measured), plus the CPU `memcpy` (~0.1 ms each) and the memory traffic.
First GPU operation after idle: 0.3–1.1 ms per frame at 30 fps	Lock the clocks for low-rate perception (`jetson_clocks` measured; a steady background load is untested), or warm the GPU before the frame arrives, the GPU analogue of keeping a CPU out of deep idle (axon's `CpuLatencyHold`).
Timestamp = first USB packet	When fusing with proprioception (joint encoders, tactile), shift the camera time back by the exposure and readout the timestamp does not include.
Consumer on an isolated CPU; no frame skipped in 9,600 frames	Pin the perception consumer away from I/O and other inference; see the RT section of `docs/hardware-verification.md` §7.

7. Limitations

- One camera, one resolution (640×480 YUYV), one frame rate (30 fps), one board. Larger frames make the H2D copy, and so the zero-copy saving, proportionally larger; a

RealSense run is planned.

- The copy baseline uses pageable host memory. Pinned memory, or mapped (zero-copy) host memory on the integrated GPU, would shorten or remove the H2D copy without axon's `NvBufSurface` pool; they were not measured.
- No ROS 2 baseline on the board (ROS 2 is not installed on it); the copy path stands in for a conventional pipeline.
- The exposure time and the true glass-to-inference latency are not measured; that needs an external light source or display in view of the camera.
- The kernel uses BT.601 full-range coefficients while the camera reports limited range. This shifts colours, not timing.
- Two runs per cell. The per-frame distributions are narrow (Figure 3), but run-to-run variation beyond two runs is not characterised.

8. Reproduce

```
cmake -S . -B build -DAXON_ENABLE_JETSON=ON -DAXON_WITH_CUDA=ON && cmake --build build -j
/usr/src/tensorrt/bin/trtexec --onnx=resnet18-v1-7.onnx --fp16 --saveEngine=resnet18_fp16.engine
E2E_WARM_GPU=1 instrumentation/jetson/e2e/run_e2e.sh build resnet18_fp16.engine 600 2 my_run
sudo jetson_clocks --store clocks.conf && sudo jetson_clocks          # locked-clock cells
E2E_WARM_GPU=1 instrumentation/jetson/e2e/run_e2e.sh build resnet18_fp16.engine 600 2 my_run_jc
sudo jetson_clocks --restore clocks.conf
python3 docs/whitepapers/jetson-camera-e2e/make_figures.py       # figures + numbers.json
```

Raw data: `instrumentation/jetson/e2e/results/{default,jetson_clocks}_{warm,nowarm}/`, one CSV row per frame (`seq,t_cap,t_pub,t_view,t_gpu,t_pre,t_inf,top1,seqlock_retries`), consumer summaries, producer logs and `env.txt` (kernel, power mode, isolated CPUs, camera format, GPU clocks, warm-up setting; the `*_nowarm` runs predate the clock record). Every number in this paper is computed by `make_figures.py` into `numbers.json`.